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Team: SEG26-33 Meraki Arts

Project: Meraki Arts Africa Website

Feature: Event Registration

Team repository: <https://github.com/Dept-of-Comp-Sci-University-of-Ghana/dcit208-client-project-2026-team-engineering-repository-seg26-33-merakiartsafrica.git>

Individual IA9 repository: <https://github.com/Dept-of-Comp-Sci-University-of-Ghana/ia9-security-privacy-and-maintainability-review-Elorm07.git>

Related team issue/PR:

B. Linked requirement and user story

Requirement= The system should allow visitors to register for upcoming Meraki Arts Africa events.

User story= As a visitor, I want to register for an upcoming event so that I can participate in Meraki Arts Africa activities.

Acceptance criteria:

- Visitors can submit their required registration details.
- The system validates the submitted information.
- A participant cannot register for the same event more than once.
- Registration is not allowed when an event has reached its capacity.

C. Data inventory

The main data is:

Data	Purpose	Sensitivity	Storage	Access	Data type
First name	Identify participant	Low	MongoDB	Participant/Admin	Synthetic
Last name	Identify participant	Low	MongoDB	Participant/Admin	Synthetic
Email	Contact participant	Medium	MongoDB	Participant/Admin	Synthetic
Phone number	Contact participant	Medium	MongoDB	Participant/Admin	Synthetic
Event ID	Identify selected event	Low	MongoDB	System/Admin	Synthetic
Registration status	Track registration	Low	MongoDB	Participant/Admin	Synthetic

D. Access-control rule

Action	Visitor/Participant	Administrator
Create registration	Yes	Yes
View own registration	Yes	Yes

View other registrations	No	Yes
Update registration	No	Yes
Delete/cancel registration	No	Yes
Manage events	No	Yes
Export registrations	No	Yes

E. Abuse case/privacy risk

Risk= Unauthorized access to participant information. An attacker could attempt to access another participant's registration by manipulating a registration ID or API request. This could expose names, email addresses, or phone numbers and would be a privacy violation.

The system should:

- Authenticate users where required.
- Authorize requests on the backend.
- Ensure a participant can only access their own records.
- Allow administrators to access registration records according to their role.
- Never rely only on frontend restrictions for authorization.

F. Failure walkthrough

Scenario= A visitor submits a valid registration while MongoDB is unavailable.

User sees= A clear error such as *"Registration could not be completed. Please try again later."*

Data preserved= The system should not create a partial registration. Existing valid records should remain unchanged.

Logged=The backend should record the database connection/request failure without logging passwords, tokens, or unnecessary personal information.

Recovery= The team checks the backend and MongoDB logs, restores database connectivity, and retries the registration. The team then verifies that a valid registration can be completed successfully.

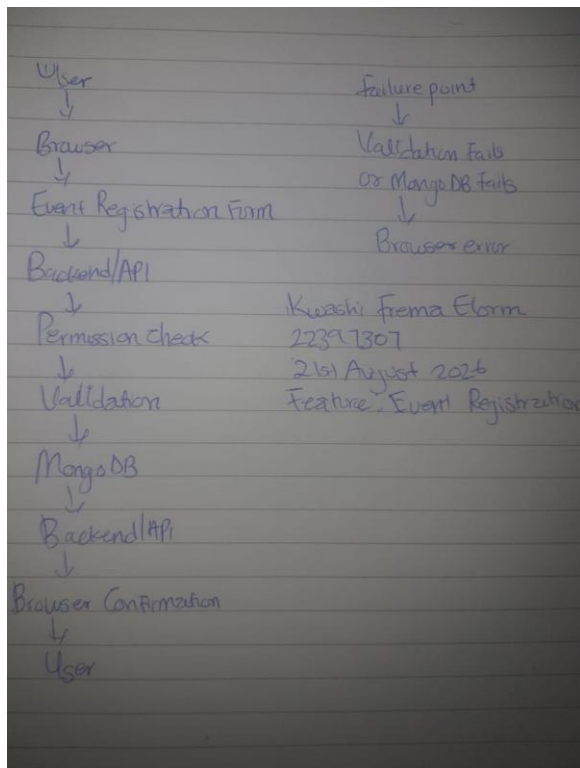
G. Verification plan

- Attempt to register with valid information.
- Attempt to register with missing required fields.
- Attempt duplicate registration for the same event.

H. Maintenance and handoff

- MongoDB connection configuration must be maintained securely.
- Production secrets must be stored as environment variables.
- Registration validation exists on the backend and should not rely only on the frontend.
- Duplicate-registration protection must remain in place.
- A future improvement could be adding email confirmation after successful registration.
- Another future improvement could be stronger authentication and role-based access control if the current implementation is limited.

I. Hand-drawn sketch



J. Oral defence video link: <https://youtu.be/WVInAEKB4Vk>

K. AI disclosure= AI was used to assist with organizing and wording parts of this deployment documentation. It helped structure the deployment plan, environment and configuration information, smoke-test scenarios, failure-handling process, rollback procedure, and oral defence script. I personally reviewed and verified the technical information included in this document, including the project repositories, selected Event Registration feature, current deployment status, technology stack, configuration requirements, and planned deployment process. I also ensured that no real secrets, passwords, API keys, or sensitive credentials were included. AI was not allowed to invent deployment evidence, claim that the feature had been implemented or deployed when it had not, create false GitHub commits, PRs, release tags, CI results, smoke-test results, or deployment URLs. Any evidence stated as unavailable reflects the actual current project status. The final responsibility for the accuracy of the submitted documentation and technical decisions remains mine.